

Replication Guide

This document lists the exact commands to reproduce **every table and figure** in the project report from a fresh clone of this repository.

0. Environment

```
git clone <this-repo-url>
cd <repo-dir>
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
```

Tested with Python 3.13.7, TensorFlow 2.20.0, Keras 3.13.2 on macOS Darwin 25.4.0. Pre-trained DNNs and processed CSVs are committed under `DNN/` and `dataset/`; no external download is required.

1. Train sklearn baselines

```
python -m ga_fairness_tester.train_sklearn_models
```

Produces `models/{adult,compas}_{lr,rf,gbm}.pkl`. Each model is trained on an 80/20 stratified split with `random_state=0`. Reported test accuracies (for sanity checking, not used in the experiment):

adult	lr	0.754
adult	rf	0.812
adult	gbm	0.840
compas	lr	0.706
compas	rf	0.682
compas	gbm	0.709

2. Run the full experimental matrix

```
python -m ga_fairness_tester.experiment \
  --datasets adult compas \
  --models dnn lr rf gbm \
  --runs 30 \
  --budget 1000 \
  --out results/results.csv
```

Configurations: 2 datasets × 4 models × |sensitive_features| × 30 seeds × 2 algorithms. Total runs: **1200** (adult has 3 sensitive features, compas has 2, so 5 sensitive features in total).

Output: `results/results.csv` with one row per run, schema:

```
algorithm, dataset, model, sensitive_feature, run_id, budget,
n_idis, idi_ratio, idi_diversity, runtime_seconds
```

Wall-clock runtime: ~1.5 hours on a 2024 MacBook Pro (M-series CPU). DNN runs dominate runtime; the three sklearn families together complete in under a minute per configuration.

3. Statistical analysis

```
python -m ga_fairness_tester.stats \
  --results results/results.csv \
  --out      results/stats_summary.csv
```

Produces `results/stats_summary.csv` with per-configuration medians, IQRs, Wilcoxon rank-sum p-values, Vargha-Delaney A12 effect sizes and verdict labels. The summary table in the report is a direct subset of these columns.

4. Figures

```
python -m ga_fairness_tester.visualise \
  --results results/results.csv \
  --out      figures/
```

Produces: - `figures/box_<dataset>_<sensitive>.png` — IDI ratio box plots, RS vs GA-FT, one panel per (dataset, sensitive feature) combination, with all four model families side by side. -

figures/conv_<dataset>_<model>_<sensitive>.png — convergence curves (unique IDIs found vs evaluations consumed) plotted as median $\pm$ IQR across 30 independent seeds per (dataset, model, sensitive) tuple.

All figures are 150 DPI matplotlib PNGs (no screenshots).

5. α/β sensitivity ablation (§5.5 of the report)

```
python -m ga_fairness_tester.ablation --seeds 10 --budget 1000
```

Sweeps $\alpha \in \{0.0, 0.3, 0.5, 0.7, 0.9, 1.0\}$ with $\beta = 1 - \alpha$ on the ADULT / GBM / gender configuration. Writes results/ablation.csv and the figures/ablation_alpha.png plot used in the report.

6. Unit tests

```
pytest tests/ -q
```

24 tests across tests/test_oracle.py, test_fitness.py, test_operators.py, test_stats.py. All pass on a clean install.

7. Determinism

All algorithms use `numpy.random.default_rng(seed)` and identical seeds for RS and GA-FT within a given run, so the trajectories are bit-exactly reproducible across machines (modulo TensorFlow non-determinism inherent to GPU operations — CPU inference is deterministic in our setup).